USE A FRESH UBUNTU!

Part 1: Research

1. What is Ansible and why is it widely used in server management?
   - **Ansible is an open source automation tool, it use widely because of its effiency**
2. What role does SSH play in how Ansible manages servers?
   - **enables Ansible to manage servers effectively without requiring any agent software to be installed on the managed nodes themselves**
3. What are the purposes of the following files?
   - `ansible.cfg`
   - **This is basically the caller in simple terms or defining the location of each files like for example the ssh private key and inventory.ini**

   - `inventory.ini`
   - **The purpose of inventory defines or manage the host of and group them so that we can use playbook correctly with the connection on inventory.**

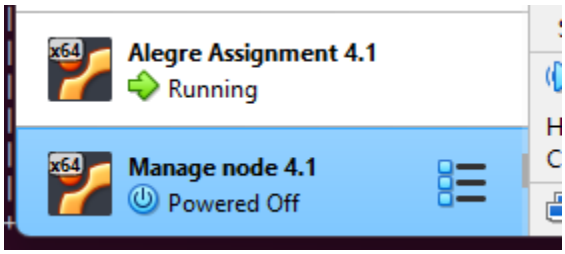
   - A playbook (`.yml` file)
   - **a YAML file that acts as a human-readable blueprint for automating IT tasks, defining what to do (tasks), where (hosts/inventory)**

Part 2: Reading & Summary

1. Read about how to install Ansible on Ubuntu from at least **one reliable online source** (official docs, blogs, tutorials).
2. Write a **short summary (5–7 sentences)** explaining:

The installation steps:

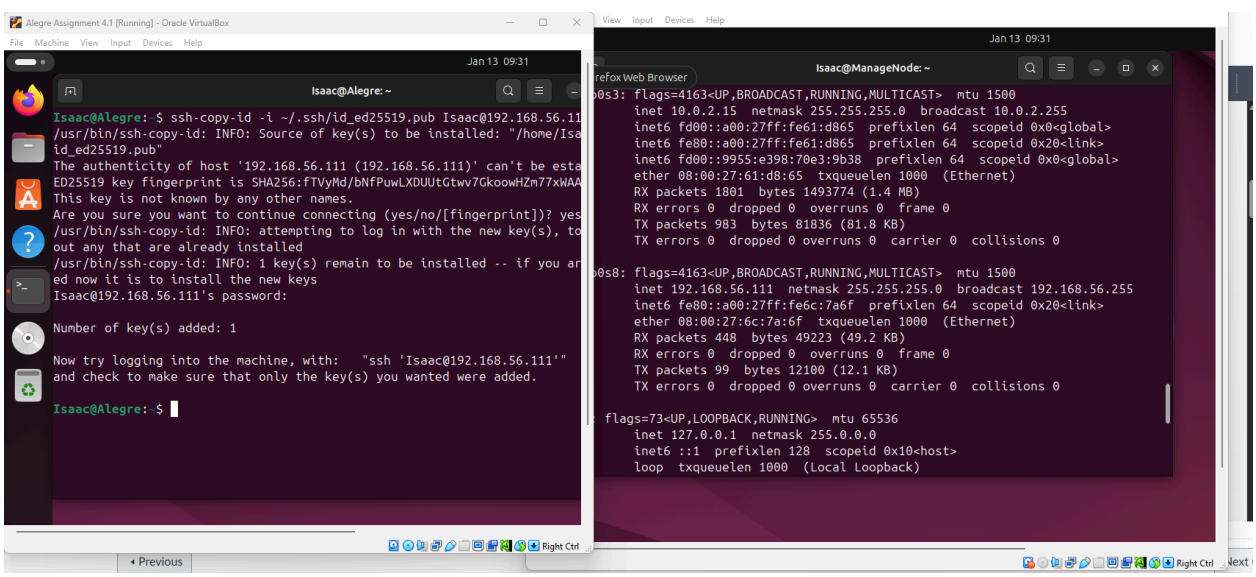
First, create 2 vm, 1 for manage node then 1 for control node



Second is generate an ssh on control node:

```
ubuntu@ubuntu:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:gVtN6fvniUeqhRMjqK4+vgN8ttVxymWioZ2Mf2Rusao ubuntu@ubuntu
The key's randomart image is:
+--[ED25519 256]--+
|          ..     |
|       . o.      |
|      . o..      |
|     . * =.      |
|.   = O S o.     |
|.. = B B ..+  .  |
| .o = + o o..o   |
|  oo . =   oo.o. |
| o=Eo.+   ...+o  |
+----[SHA256]-----+
ubuntu@ubuntu:~$
```

Third is copy your ssh key into your manage node



The purpose of an inventory

- The purpose of inventory defines or manage the host of and group them so that we can use playbook correctly with the connection on inventory.

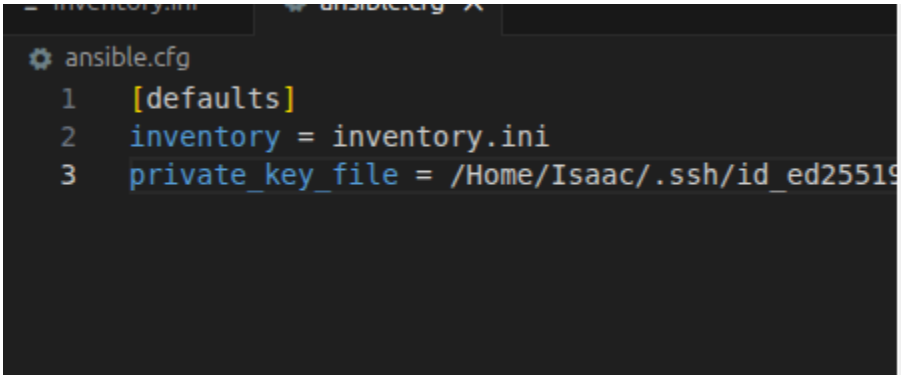
The purpose of a configuration file

- This is basically the caller in simple terms or defining the location of each files like for example the ssh private key and inventory.ini

Part 3: Organizing

1. Create a simple **draft version** of:
    - A sample `inventory.ini` (with at least 2 pretend server IPs).

    - A sample `ansible.cfg` file with at least 2 meaningful settings (e.g., inventory and remote_user).



Part 4: Install an useful application used in Development, do not use applications that we have installed before.

File Edit Selection View Go ··· ansible

inventory.ini | ansible.cfg | playbook.yml

playbook.yml

```yaml
- hosts: all
  become: true
  tasks:
    - name: Update apt cache
      ansible.builtin.apt:
        update_cache: yes

    - name: Install Node.js and npm
      ansible.builtin.apt:
        name:
          - nodejs
          - npm
        state: present

    - name: Install Firebase CLI globally using npm
      ansible.builtin.npm:
        name: firebase-tools
        global: true
        state: present
```

Result:

```yaml
tasks:
  - name: Update apt cache
    ansible.builtin.apt:
      update_cache: yes

  - name: Install Node.js and npm
    ansible.builtin.apt:
      name:
        - nodejs
        - npm
      state: present

  - name: Install Firebase CLI globally using npm
    ansible.builtin.npm:
      name: firebase-tools
      global: true
      state: present
```

```
Isaac@Alegre:~/ansible$ ansible-playbook ./playbook.yml -K
TASK [Gathering Facts] ********************************************************************************
ok: [192.168.56.111]

TASK [Update apt cache] *******************************************************************************
changed: [192.168.56.111]

TASK [Install Node.js and npm] ************************************************************************
ok: [192.168.56.111]

TASK [Install Firebase CLI globally using npm] ********************************************************
changed: [192.168.56.111]

PLAY RECAP ********************************************************************************************
192.168.56.111             : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

Isaac@Alegre:~/ansible$
```

Verify:

```
Isaac@ManageNode:~$ nodejs --version
v18.19.1
Isaac@ManageNode:~$ nodejs --version
v18.19.1
Isaac@ManageNode:~$ firebase --version
Firebase CLI v15.2.1 is incompatible with Node.js v18.19.1 Please upgra
s to version >=20.0.0 || >=22.0.0 || >=24.0.0
Isaac@ManageNode:~$ npm --version
9.2.0
Isaac@ManageNode:~$
```